

Qualifying Exam in Microeconomics
Arizona State University
August 2021

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. Consider this prisoners' dilemma stage game.

		Player t (young)	
		C	D
Player $t - 1$ (old)	C	2, 2	-1, 3
	D	3, -1	0, 0

Figure 1: Overlapping-generations stage game

This game is to be played by a *countable infinity* of players $i = 0, 1, 2, 3, \dots$ with overlapping generations. Player 0 lives only at date 1. Every other player $i = 1, 2, 3, \dots$ lives two dates, date i and date $i + 1$. At each date $t \geq 1$, the two players alive at that date—player t , the *young* player, and player $t - 1$, the *old* player—simultaneously choose C or D after observing the entire history of past play. Every player discounts future utility at rate $\delta \in (0, 1)$.

- (a) What is the strategy set of a player i in this game? (Throughout, please confine your attention to “pure” strategies.)
- (b) Show that, if the discount factor δ is large enough, then there is a Nash equilibrium that supports the outcome “each player when young chooses C and each player when old chooses D .” Explain carefully how the strategy profile you identify is a Nash equilibrium and supports this outcome for high-enough δ . How low can the discount factor be?
- (c) Can the outcome from part (b) be supported as a *subgame-perfect* Nash equilibrium for large-enough δ ? If it can, prove it; if it can't, explain why subgame perfection must fail when this outcome is supported as a Nash equilibrium.
- (d) You confirmed in part (b) that a Nash equilibrium of the infinite-period game exists. Can you invoke the Debreu-Fan-Glicksberg theorem on the existence of Nash equilibrium to explain why this infinite-period game has a Nash equilibrium? *Briefly* explain why or why not.

2. Consider a principal-agent problem with moral hazard, a continuum of effort levels $a \geq 0$ and two output levels $x \in \{x_\ell, x_h\}$, $0 < x_\ell < x_h$. The principal is risk neutral, and the agent is strictly risk averse with utility function that is additively separable in income and effort, given by $\log(\xi + w) - \psi(a)$ for wage-effort pairs (w, a) , with $\psi(0) = \psi'(0) = 0$, $\psi' > 0$ for $a > 0$, and $\psi'' \geq 0$, and where $\xi > 0$ is the agent's initial wealth (which is a parameter of the problem and is publicly observable). The probability of x_h when the agent exerts effort a is $p(a)$, and thus the probability of x_ℓ is $1 - p(a)$, where $p(0) \in (0, 1)$, $p' > 0$ and $p'' < 0$. The agent's reservation utility is $\log(\xi + \bar{w})$, where $\bar{w}$ is the wage the agent can obtain with certainty in her outside option (also publicly observable), which we assume is large enough so that the nonnegativity constraint (due to log utility) on initial wealth plus wage for each output realization can be ignored.

- (a) Set up the principal's contracting problem, and explain the two steps you would follow to solve the problem. Is the first-order approach valid? Justify your answer.
- (b) For each effort level $a \geq 0$, obtain the compensation scheme (in terms of a utility level for each realization of x) that minimizes the principal's expected cost of inducing the agent to exert that effort level. Use any theoretical result you know that may allow you to simplify the algebra. Explain the properties of the cost-minimizing compensation scheme.
- (c) Show that the principal weakly prefers a poorer agent (that is, one with a lower ξ).

3. A worker privately observes her type $t \in [0, 1]$, drawn randomly using the uniform distribution. After observing her type the worker may acquire education $e \geq 0$ at a cost $\frac{e}{t}$.

Each firm values a type- t worker with education e by $f(t, e) = t^{1/2}$. A worker with type t and education e that receives a wage $w(e)$ has payoff

$$w(e) - \frac{e}{t}.$$

- (a) Please state whether the following statement is true or false and briefly explain your reasons.

An equilibrium requires that for every e the wage $w(e)$ equals the firms' expected valuation of those workers who acquire education e .

- (b) Use the FOC of the worker's problem and part (a) to derive a differential equation that characterizes the wage function in a separating equilibrium, if it exists and $w(e)$ is differentiable. (You need not solve the differential equation.)
- (c) Suppose $w(e) = (3e)^{1/3}$. What proportion of a type t value to the firm is spent on the costs of education?

4. A person has expected-utility preferences over wealth distributions with compact support in $\mathbb{R}$ represented by the quadratic vN-M utility function u on all of $\mathbb{R}$ given by

$$u(z) = az - \frac{1}{2}z^2, \quad (1)$$

where $a > 0$.

Parts (a)-(b) are preliminary; the substance of the question is in parts (c)-(d).

- (a) If the wealth distribution is given by a cumulative distribution function (cdf) F with bounded support, show that expected utility, $\int u(z)dF(z)$, can be written as a function of the mean wealth, $\mu_z = \int zdF(z)$, and the variance of wealth, $\sigma_z^2 = \int z^2dF(z) - \mu_z^2$.

The person has initial wealth $w \in (0, a)$ which must be put in one of two assets: a safe asset (m) and a risky asset (α), both measured in the same units as w ; the budget constraint must therefore hold with an equality, $w = m + \alpha$. *Each asset holding is constrained to be nonnegative:* $(\alpha, m) \geq 0$.

The safe asset has a rate of return of 0 and the risky asset has an *ex post* rate of return of x . The distribution of the risky asset's return is given by a nondegenerate cdf G with bounded support in $\mathbb{R}$. (By nondegenerate, we mean that it does not put probability 1 on a single point, that is, the asset is indeed risky.)

The person's problem is

$$\begin{aligned} \max_{(\alpha, m) \geq 0} \quad & \int u(\alpha(1+x) + m) dG(x) \\ \text{subject to} \quad & m + \alpha = w, \end{aligned} \quad (2)$$

with u given by (1).

- (b) Write the objective in (2) as a function of the mean return on the risky asset, $\mu_x = \int xdG(x)$, and the variance of the return on the risky asset, $\sigma_x^2 = \int x^2dG(x) - \mu_x^2$. (Suggestion: use part (a) and write μ_z and σ_z^2 as functions of μ_x and σ_x^2 .)
- (c) Solve for the person's demand for the *risky* asset (α) as a function of w , μ_x , and σ_x , *carefully justifying your steps*. Remember that the only assumptions on G is that it is nondegenerate and it has bounded support.
- (d) How does an increase in w affect the person's demand for the risky asset? What aspect(s) of preferences explains your finding?